



Occurrence, molecular characterization and antimicrobial-resistance pattern of *Staphylococcus* species isolates from buck semen

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Abstract

Staphylococcus aureus is one of the most prevalent pathogens, and a causative agent of a variety of infections in humans and animals. Most studies concentrated on characterization of staphylococcus isolates and its antimicrobial resistance from various illness of veterinary importance, but there is no specific study that is available on isolates from reproductive tract of small ruminants and especially its semen. Hence, in the current study, a total of 48 semen samples were collected from healthy bucks of different breeds to investigate the occurrence of *S. aureus*. Antimicrobial resistance and virulence of the *Staphylococcus* isolates were determined to assess the adverse effects of them on buck fertility. The bacterial isolates were tentatively confirmed as *Staphylococcus* spp. based on the Gram's staining, growth on Mannitol salt agar and catalase test. Overall, 75% ($n=36$) of the samples were positive for *Staphylococcus* spp. from the total 48 buck semen ejaculates from different breeds and among them 23 (63.89%) were coagulase-negative (CoNS) and 13 (36.11%) were coagulase-positive *Staphylococcus* (CoPS) strains. The species identified by molecular characterization are *S. aureus*, *S. chromogenes*, *S. haemolyticus*, *S. sciuri*, *S. simulans*, and *S. epidermidis* from buck semen. Further, these isolates exhibited varying degrees of multidrug resistance genotypically as well as phenotypically. The presence of antibiotic resistance and virulence genes may pose a potential threat to reproductive health of animals, the animal handlers and livestock keepers, while simultaneously highlighting the need for vigilant monitoring of these isolates at the time of semen cryopreservation.

Keywords Buck semen · *Staphylococcus* spp. · CoNS · Methicillin resistance · Vancomycin resistance · Reproductive health

Introduction

Bacteria are present in every ejaculate, and the aim of obtaining sterile semen is obviously non-achievable task. Bacteria can gain access to the semen as a result of bacteremia, due to local infections in parts of the genital tract, and inflammation or trauma of the urinary tract and the preputial

cavity. Next source of semen bacterial contamination is the saprophytic flora of the prepuce in healthy semen donors, fecal bacteria, bacteria from substances of animal origin that are added to the semen diluents or from equipment and materials or workers and airborne bacteria (Gangwar et al. 2020).

Staphylococcus aureus, a Gram-positive bacterium, is a causative agent of a variety of infections in humans and animals (McMillan et al. 2016). In animals, *S. aureus* is associated with mastitis, one of the most cost-intensive diseases in the dairy industry. Mastitis is an infectious disease responsible for significant financial losses to dairy and food farmers worldwide (Xing et al. 2016). Methicillin-resistant *S. aureus* (MRSA) was first reported in the United Kingdom, now it has become a public threat to human health, and various hospital-associated MRSA (HA-MRSA) clones have been disseminated globally (Monecke et al. 2011). Since the 1990s, community-associated MRSA (CA-MRSA) has emerged

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as a serious health problem in entire world (DeLeo et al. 2010). Livestock-associated MRSA (LA-MRSA) clones, such as LA-MRSA ST398 may be transmitted to humans, and have posed public health concerns (Fitzgerald 2012). Therefore, it is imperative to perform surveillance at the interface between human and animal hosts to explore human health risks (Cuny et al. 2015). Nowadays, the emergence of multidrug resistant (MDR) strains poses several challenges to the clinical facilities (Hammad et al. 2012). In particular, the increasing prevalence of antimicrobial-resistant *S. aureus* serves as a threat to the healthcare system (Rhee and Woo 2010; Argudín et al. 2012). On the other hand, most of *S. aureus* strains are able to produce a large number of virulence factors, including staphylococcal enterotoxins (SEs), exfoliative toxin (ET) and toxic shock syndrome toxin-1 (TSST-1). Moreover, the production of SEs is particularly significant, as the ingestion of the pre-formed toxins is a major cause of foodborne poisoning worldwide (Basanisi et al. 2017).

These pathogens, in one way or other, may infect inseminated females or contribute to a rapid deterioration in sperm quality (Al-Kass et al. 2019). Nevertheless, scientific evidence based on published data concerning the major risk pathogens or bacterial contamination of the germplasm used in goat AI operations in India has been lacking until recently. The aim of the present study was to investigate the occurrence and perform molecular characterization of *Staphylococcus* isolates from the semen samples of bucks. Furthermore, we evaluated the antimicrobial susceptibility, resistance genes, and virulence genes of these isolates. Moreover, this type of work has not been carried out in goats. Findings of this study throw light in the darkness of pathogenicity of *Staphylococcus* spp. present in buck semen.

Materials and methods

Semen collection for bacteriological study

Total 48 semen ejaculates were collected from different goat breeds (Barbari, $n = 12$; Jamunapari, $n = 12$; Jakhrana, $n = 12$; Bundelkhandi, $n = 12$) with good health condition, which were randomly selected during the breeding season, and maintained under semi-intensive system of rearing with same managerial conditions throughout this study. The whole experiment is carried out as per the guidelines of the Institute Animal Ethics Committee (IAEC).

Microbiological evaluation of semen

The fresh semen samples were inoculated into nutrient broth separately, and incubated at 37 °C overnight. The broth culture was inoculated on the nutrient agar and mannitol salt

agar according to the method described by Mishra et al. (2014).

For identification of bacteria (*Staphylococcus aureus*), Gram's stain was performed according to the method described by Merchant and Packer (1967) (refer to Fig. 1 in supplementary materials). For catalase test, a loopful of fresh culture was mixed with 3% Hydrogen peroxide in a clean glass slide, and positive reaction is characterized by effervescence due to evolution of oxygen (refer to Fig. 2 in supplementary materials). For identifying coagulase-positive staphylococci, coagulase test was performed using rabbit plasma as per the standard protocol. For coagulase test, a well-isolated staphylococcal colony is to be emulsified in a drop of water on a clean and grease-free glass slide with minimum of spreading. A flamed and cooled straight inoculating wire is dipped into the undiluted plasma at room temperature, and the adhering traces of plasma (not a loopful) is mixed into the staphylococcal suspension on the slide. In positive case, a coarse clumping of cocci is visible to the naked eye within 10 s (refer to Fig. 3 in supplementary materials).

Molecular characterization of *Staphylococcus* spp.

Molecular characterization was done in two steps. Initially, semen samples were subjected to DNA extraction using Nucleo-pore Fungus Bacteria genomic DNA kit (Cat#NP-7006D) following the manufacturer's instructions and further checked for its quantity and quality using Quantus™ fluorometer using the Quanti Fluor® ONEdsDNA system (E4871) following the manufacturer's instructions. Multiplex PCR was performed as described by Shome et al. (2011) incorporating the oligonucleotide sequences along with primers for *nuc* gene standardized and developed by Brakstad et al. (1992) were included in the current study (Table 1). Finally, the semen samples were bacteriologically cultured for isolation of *Staphylococcus* species and 36 isolates which were obtained were also screened by multiplex PCR as mentioned above.

The primers are used in the specified concentration as mentioned above along with 2 × Emerald GT amp master mix (TAKARA, Japan), along with 1 µl template DNA (equivalent to approx. 2–3 ng of bacterial genomic DNA) to make up the total volume of reaction to 25 µl. The PCR was setup with the following thermal conditions:

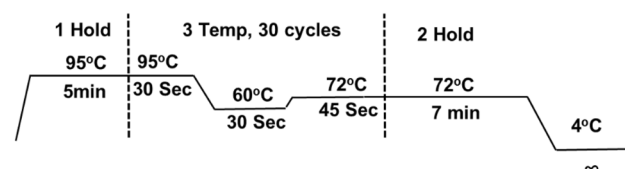


Table 1 List of oligonucleotides used along with details of concentration, amplicon size and number of various species-wise *Staphylococci* isolates obtained after typing by multiplex PCR in buck semen

Name of gene	Oligonucleotide sequence 5' → 3'	Size (bp)	Conc used (μM)	Species	No. of isolates screened	Reference
23srRNA	F-AGCGAGTCTGAATAGGGCGTTT R-CCCATCACAGCTCAGCCTTAAC	894	0.25	<i>S. aureus</i>	13	Shome et al. 2011
Nuc	F- GCGATTGATGGTGATACGGTT R- AGCCAAGCCTTGACGAATAAAGC	278	0.8			Brakstad et al. 1992
SodA	F-GCGTACCAGAAGATAAAACAACTC R-CATTATTTACAACGAGCCATGC	222	0.2	<i>S. chromogenes</i>	11	Shome et al. 2011
SodA	F-CAAATTAATTTCTGCAGTTGAGG R-AGAGCCCCATTGTTCTTTGA	214	0.2	<i>S. haemolyticus</i>	2	
Gap	F-GATTCCGCGTAAACGGTAGAG R-CATCATTTAATACTTTAGCCATTG	306	0.2	<i>S. sciuri</i>	3	
Gap	F-AGCTTCGTTTACTTCTTCGATTGT R-AAAAGCACAAGCTCACATTGAC	472	0.2	<i>S. simulans</i>	6	
Rdr	F-AAGAGCGTGGAGAAAAGTATCAAG R-TCGATACCATCAAAAAGTTGG	130	0.2	<i>S. epidermidis</i>	1	
Total number of various <i>Staphylococcus</i> spp. isolates screened					36	

Antimicrobial-resistance studies for *Staphylococcus* spp. isolates from buck semen

The confirmed coagulase-negative *Staphylococci* (CoNS) and coagulase-positive *Staphylococci* (CoPS) isolates were then subjected to β-Lactamase production and Methicillin-resistant *Staphylococcus aureus* (MRSA) typing using both phenotypic and genotypic tests. All the antimicrobial tests were performed according to CLSI standards (CLSI 2013) and their break-points.

β-Lactamase production in *Staphylococcus* spp. by Penicillin zone edge test

This test was performed using disk diffusion test by placing penicillin (10 U) disk, and incubating at 35 °C for overnight (16–18 h) incubation. The tests are interpreted by observing the edges of inhibition zones which appear as sharp edge (cliff-like) as β-Lactamase positive and fuzzy edge (beach-like) as β-Lactamase negative (refer to Fig. 4 in supplementary materials).

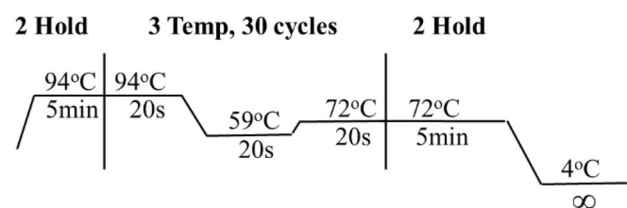
Methicillin-resistant *Staphylococcus aureus* (MRSA) typing using phenotypic tests

The phenotypic MRSA test was performed using the disk diffusion test employing cefoxitin disk (30 μg) for *S. aureus*, and CoNS according to CLSI standards. The inoculum was prepared, and spread on the Mueller Hinton agar (Oxoid, Thermofisher), and the antibiotic disk was placed over the inoculated plate after 15 min, and incubated at 35 °C for

18 h (*S. aureus*), and 24 h (for CoNS). The cefoxitin disk (30 μg) with a minimum zone of inhibition (ZoI) of 22 mm and above is indicative of MRSA, and less than 22 mm is not indicative of MRSA (refer to Fig. 5 in supplementary materials). Similarly, for CoNS (except *S. pseudointermedius* and *S. schleiferi*) the ZoI should be ≥ 25 mm for MRSA.

Methicillin-resistant *Staphylococcus aureus* (MRSA) typing using molecular methods

All the CoPS and CoNS isolates were subjected to genotypic MRSA detection using the genes viz., *mecA* and *mecC* (Table 2). The primers used and their thermal cycling conditions are mentioned below.



The primers were used at a volume of 1 μl (10 pMol/μl) each of the forward and reverse primer along with 1 μl template DNA (equivalent to approx. 2–3 ng of bacterial genomic DNA), in 12.5 μl of 2× Emerald GT amp master mix (TAKARA, Japan), and nuclease-free water added in rest of volume to make up the total volume of reaction to 25 μl. The PCR was setup for both the MRSA genes with the same thermal conditions as illustrated above.

Table 2 Details of oligonucleotides used for MRSA and VRSA genotyping PCR

S. no	Name of the MRSA gene	Oligonucleotides (5' → 3')	Product length	References
1	<i>mecA</i>	<i>mecAF</i> : TCCAGATTACAACCTCACCAGG <i>mecAR</i> : CCACTTCATATCTTGTAACG	162 bp	Oliveira and Lencastre (2002)
2	<i>mecC</i>	<i>mecCF</i> : GAAAAAAGGCTTAGAACGCCTC <i>mecCR</i> : GAAGATCTTTCCGTTTTCAGC	138 bp	Stegger et al. (2012)
3	<i>vanA</i>	<i>vanAF</i> : GGG AAA ACG ACA ATT GC <i>vanAR</i> : GTA CAA TGC GGC CGT TA	732 bp	Bhatt et al. (2015)
4	<i>vanB</i>	<i>vanBF</i> : ACGGAATGGGAAGCCGA <i>vanBR</i> : TGCACCCGATTTCGTTC	647 bp	

Vancomycin-resistant *Staphylococcus aureus* (VRSA) detection by genotypic test

All the isolates of *Staphylococcus* ($n = 36$) were subjected to VRSA detection using two sets of genes viz., *vanA* and *vanB* (Table 2). The details of the primers and their conditions are described in Table 3. The concentration and volume of the primers and reagents including the thermal conditions were similar to the previous section.

Results

Isolation of *Staphylococcus* from Buck semen

The bacterial isolates were tentatively confirmed as *Staphylococcus* isolates based on the Gram's staining, growth on mannitol salt agar (refer to Fig. 1 in supplementary materials) and positive catalase test (Fig. 2, supplementary materials). From the total 48 buck semen ejaculates from different breeds, 36 (75%) semen samples were positive for *Staphylococcus* spp. Out of 36 staphylococcal isolates, 23 (63.89%) were coagulase-negative *Staphylococcus* (CoNS) strains and 13 (36.11%) were coagulase-positive *Staphylococcus* (CoPS) strains; however, only 27.78% were phenotypically coagulase positive based on the slide coagulase test.

Molecular characterization of *Staphylococcus* spp.

At first step, DNA extracted from buck semen was directly subjected to multiplex PCR for detect the presence of various *Staphylococcus* spp. (Fig. 1). Simultaneously semen samples were cultured bacteriologically to isolate various strains of *Staphylococcus* spp., which were further subjected to species level identification by molecular characterization using conventional multiplex PCR. In the current study, we isolated *S. aureus*, *S. chromogenes*, *S. haemolyticus*, *S. sciuri*, *S. simulans* and *S. epidermidis* in the buck semen as given in the Table 1.

Phenotypic AMR typing

Detection of β -Lactamase-producing *Staphylococcus* isolates by penicillin zone edge test

The *Staphylococcus* isolates were initially screened for the beta-lactamase activity using the penicillin zone edge test (Fig. 4, supplementary materials). Out of the 36 isolates tested, 7 (19.44%) produced cliff-like zone edges suggestive of β -Lactamase-positive isolates. Among the β -Lactamase-producing *Staphylococcus* isolates, 2 were coagulase positive and 5 were coagulase negative.

Table 3 Occurrence of Staphylococcosis in buck semen

Breed	Total no of semen ejaculates tested	Semen ejaculates positive for staphylococcus	<i>Staphylococcus</i> sp. obtained based on multiplex PCR of semen samples
Barbari	12	8 (66.67%)	<i>S. aureus</i> (3/12, 25%), <i>S. epidermidis</i> (1/12, 8.33%), <i>S. simulans</i> (2/12, 16.67%), <i>S. chromogenes</i> (2/12, 16.67%)
Jamunapari	12	10 (83.33%)	<i>S. aureus</i> (4/12, 33.33%), <i>S. haemolyticus</i> (1/12, 8.33%), <i>S. sciuri</i> (1/12, 8.33%), <i>S. simulans</i> (2/12, 16.67%), <i>S. chromogenes</i> (2/12, 16.67%)
Jakhana	12	9 (75%)	<i>S. aureus</i> (3/12, 25%), <i>S. haemolyticus</i> (1/12, 8.33%), <i>S. sciuri</i> (1/12, 8.33%), <i>S. simulans</i> (2/12, 16.67%), <i>S. chromogenes</i> (2/12, 16.67%)
Bundelkhandi	12	9 (75%)	<i>S. aureus</i> (3/12, 25%), <i>S. sciuri</i> (1/12, 8.33%), <i>S. chromogenes</i> (5/12, 41.67%)
Total	48	36 (75%)	

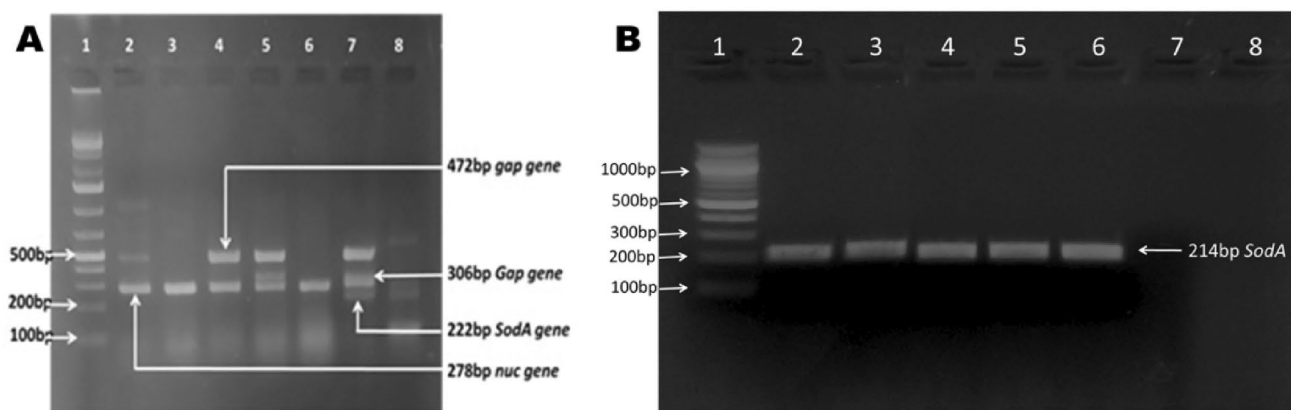


Fig. 1 Detection of *Staphylococcus* species from buck semen and isolates using conventional PCR. **A** Lanes, 1: 100 bp DNA ladder, 2–7: *nuc* gene (278 bp), 4–5: *gap* gene (472 bp), 5 and 7: *gap* gene (306 bp), 7: *sodA* (222 bp) gene, 8: no template control; **B** *GapA*

(214 bp) for detection of *S. haemolyticus* isolates, Lanes, 1: 100 bp Ladder, 2–3: unknown isolates positive for *GapA* gene (214 bp), 4–6: positive controls for *GapA* gene (214 bp), 7: no template control

Table 4 Phenotypic MRSA typing of CoPS and CoNS isolates and genotyping of MRSA-VRSA isolates by PCR

S. no	Type of staphylococcus	Coagulase positive		Coagulase negative		MRSA typing		VRSA typing	
		MRSA positive	MRSA negative	MRSA positive	MRSA negative	<i>mecA</i> +	<i>mecC</i> +	<i>vanA</i> +	<i>VanB</i> +
1	<i>S. aureus</i> (n = 13)	0	10	2	1	5	4	4	3
2	CoNS (n = 23)	–	–	4	19	5	2	5	6

Phenotypic detection of MRSA isolates

MRSA typing of the isolates were conducted and the results are summarized in the Table 4 and depicted in Fig. 4. Out of the 36 isolates, 23 (63.89%) were coagulase-negative strains and 10 were coagulase-positive (27.78%) strains. However, among the coagulase-positive *S. aureus* strains, few were phenotypically coagulase negative. From the total isolates tested ($n = 36$), 6 isolates (16.67%) were positive for MRSA by phenotypic test. Among the coagulase-positive isolates, none of them tested positive for MRSA, while all the MRSA positive isolates were coagulase negative. Two (20.0%) of the phenotypically coagulase-negative *S. aureus* ($n = 10$) and 4 (15.38%) of the CoNS ($n = 26$) were tested positive for MRSA. Among the 5 coagulase-negative β -Lactamase-producing *Staphylococcus* isolates screened (in previous section), 2 were also MRSA positive.

Genotypic methicillin-resistance and vancomycin-resistance typing

Among the methicillin-resistant isolates, it may be either *S. aureus* or CoNS and hence those that were resistant, can be methicillin-resistant *S. aureus* (MRSA) or methicillin-resistant coagulase-negative *Staphylococcus* (MRCoNS).

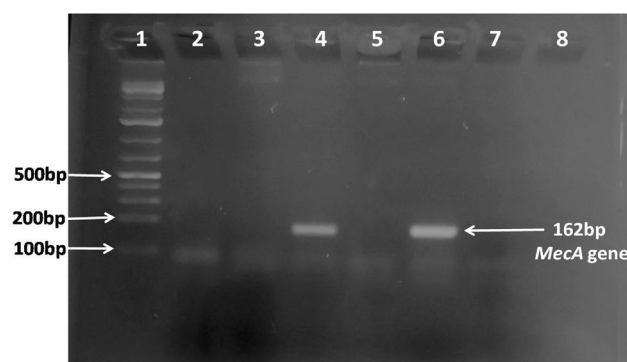


Fig. 2 MRSA genotyping based on *mecA* gene conventional PCR. Lane 1: DNA Ladder, Lane 2–5: unknown (Lane 4: positive-*mec A* gene-162 bp Isolate no. 293 Goat semen), Lane 6: positive control, Lane 7: no template control

A total of 13 *S. aureus* and 23 CoNS were tested for methicillin resistance using the *mecA* (Fig. 2) and *mecC* gene (Fig. 3) conventional PCRs. Of the 13 isolates of *S. aureus*, 5 were positive for *mecA*, and 4 were positive for *mecC*. Therefore, overall, the MRSA isolates were 5 as per the *mecA* gene which was on the higher side. As far as CoNS is concerned, 5 isolates and 2 isolates were detected for *mecA* and *mecC*, respectively, and the overall MRCoNS

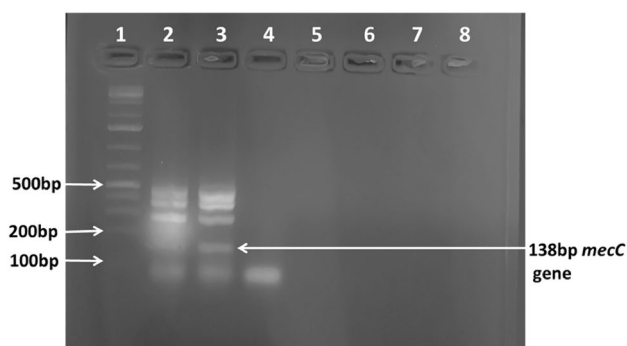


Fig. 3 MRSA genotyping based on *mecC* gene conventional PCR. Lane 1: DNA Ladder, Lane 2–3: unknown (positive for *mecC* gene-138 bp, Isolate no. 293 Goat semen), Lane 4: no template control

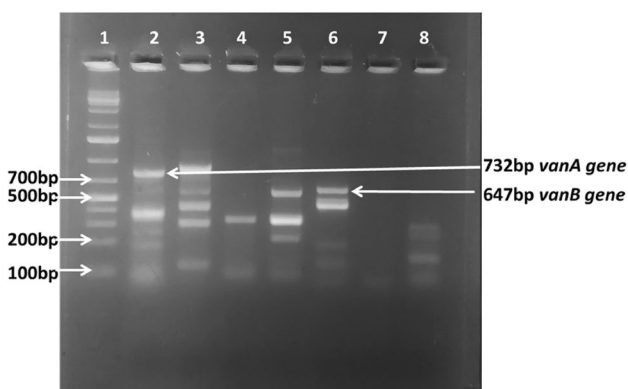


Fig. 4 VRSA genotyping based on *vanA/vanB* gene conventional PCR. Lane 1: 100 bp DNA ladder, Lane 2 and 3: *vanA* gene (732 bp), Lane 5 and 6: *vanB* gene (647 bp), Lane 7 and 8: no template controls for *vanA* and *vanB*, respectively

isolates were 5 as per the *mecA* which was on the higher side (Table 4).

Among the *S. aureus* isolates, 4 and 3 isolates were found positive for *vanA* and *vanB* genes (vancomycin resistant), respectively whereas, among the CoNS, 5 and 6 isolates were positive, respectively (Table 4 and Fig. 4).

Discussion

Infections of the male genitourinary tract may contribute to infertility by adversely affecting sperm function, causing inflammatory disorder, anatomical obstruction, alarmingly initiating leucocyte response with its concomitant oxidative stress (Gangwar et al. 2020). The effects of these conditions may be sperm damage, elevated leucocyte response, poor motility and immature sperms. Acute and chronic infections and consequent inflammation in the male reproductive system may compromise the sperm cell

function and the whole spermatogenic process (Sanocka-Maciejewska et al. 2005). The final effect usually is poor sperm quality and hence male infertility (Gangwar et al. 2021). In the present study, we investigated the occurrence and performed molecular characterization of *Staphylococcus* spp. isolated from buck semen samples to facilitate better understanding of its colonization in male genital tract and threat associated to it in buck fertility. The current study was conducted to assess the presence of *Staphylococcus* species in the semen of bucks which were used for breeding. The semen analysis showed that the CoNS isolates were more in number (47.92%) compared to CoPS (27.08%). Among the *Staphylococcus* spp. isolates, both coagulase- and non-coagulase-producing bacteria are of equal importance with the growing pathogenic nature of the later (Zong et al. 2011). The CoNS are opportunistic in nature with all the pathogenic elements (Nasaj et al. 2020), and moreover, the MRCoNS isolates are equally important in creating AMR hazard due to the harboring of *SCCmec* elements in clinical isolates (Zong et al. 2011).

Staphylococcus aureus can colonize the reproductive tract affecting its functions (Kačániová et al. 2020) and the reproductive potential of spermatozoa (Sanocka-Maciejewska et al. 2005). Most strains isolated from bull semen were *S. aureus* and *S. epidermidis*, and the metabolic products derived from these species were reported to have deleterious effects on the acrosomal integrity and motility (Meyer et al. 1980). Earlier, a study conducted on ejaculates of ram semen on bacteriological quality yielded *Staphylococcus aureus* and *Staphylococcus epidermidis* along with *E. coli*, *Enterobacter* spp., and *Proteus mirabilis* in 97% of the samples when stored at 15 °C (Yániz et al. 2010). In buffaloes, a similar study was conducted in the past revealed 11 different bacterial species in its semen including *S. aureus*, *S. intermedius*, *S. epidermidis* and others with the predominant bacteria being *S. aureus*. *S. aureus* affects the sperm function by hijacking the metabolic processes (Boryczko, 1982), thereby reducing its viability. We found that the occurrence of *S. aureus* strains (Table 1) was 75%, which was higher as compared to the previous reports (Liu et al. 2018). In the present study, *S. aureus*, *S. chromogenes*, *S. haemolyticus*, *S. sciuri*, *S. simulans*, and *S. epidermidis* were isolated from the buck semen (Table 2). The influence of gram-positive uropathogenic bacteria on sperm morphology and function has been poorly investigated until now. A previous finding reported that aerobic cocci are present in about 50% of semen samples of male partners in infertile couples (Mehta et al. 2002). One of the most frequently isolated microorganisms from male animals with genital tract infections or semen contamination is *Escherichia coli* (Diemer et al. 2003) and *Staphylococcus* spp. (Enwurua et al. 2016). In the current study, besides *S. aureus*, we

have obtained considerable number of isolates of *S. chromogenes*, *S. haemolyticus*, *S. simulans* and *S. sciuri* based on the multiplex PCR.

Staphylococcus aureus is among the most versatile and successful of the human pathogens. It has the ability to cause a variety of infections in numerous ecological niches within the host. It colonizes the nares, axillae, vagina, pharynx, semen or damaged skin surfaces and causes a variety of suppurative (pus-forming) infections and toxicoses. Besides this, *S. aureus* is arguably the dominant organism implicated in primary infertility, among males and females alike (Momoh et al. 2011). *S. aureus* has been observed as causative organism accounting for 68.2% of seminal fluid infections (Emokpae et al. 2009). This is consistent with that reported by Okon et al. (2005), where *S. aureus* was isolated from 62.5% of the seminal fluids. *S. aureus* has also been reported to be commonly isolated microorganism from cervical samples (Okonofua et al. 1995). Huwe et al. (1998) studied the influence of different uropathogenic microorganisms on human sperm motility parameters by means of CASA, and reported that *S. aureus* retards the sperm motility. Similar study was also done previously by Liu et al. (2002) where significant decrease in sperm motility was found when spermatozoa were co-incubated with *S. aureus*. The reason behind the retarded motility is due to the inhibition of Mg^{2+} ATPase activity of spermatozoa (Gupta and Prabha 2012). Some authors have suggested that direct interaction between bacteria and spermatozoa facilitates immobilization of spermatozoa (Erbengi 1993; Diemer et al. 1996; Nunez-Calonge et al. 1998), while others have reported evidence for soluble spermicidal factor produced and secreted by bacteria in the extracellular medium.

AMR is a very important aspect that requires immediate attention in any farm set up. Hence, in the current study, we screened all the isolates of *Staphylococcus* from buck semen for beta-lactamase production, methicillin resistance and vancomycin resistance. Among the coagulase-positive isolates, none of them was MRSA positive, which was an interesting finding. On the other hand, among coagulase-negative *S. aureus* isolates, a total of 6 isolates were found MRSA positive based on the cefoxitin disk diffusion phenotypic test. Similar patterns of MRSA in CoNS have been earlier reported in many instances (Poulsen et al. 2003; Otto 2013; Ruiz-Ripa et al. 2020; Li et al. 2021) and the horizontal transfer of AMR genes from less pathogenic *S. epidermidis* to *S. aureus* has been identified in certain studies during co-colonization (Otto 2013, 2020; Li et al. 2021). But the genotypic test has revealed a totally different picture with good number of *S. aureus* (CoPS) carrying both *mecA* and its homologue *mecC* gene, besides the CoNS having equal number of MRSA isolates by the same test. Usually, the genes for resistance and other pathogenic traits of *Staphylococcus aureus* are present in the mobile

genetic elements such as pathogenicity islands, plasmids, transposons and staphylococcal chromosomal cassettes (SCCs) which may be exchanged in nature through various phenomena like conjugation, phage-aided transduction and transformation (Malachowa and DeLeo 2010; Alibayov et al. 2014; Haaber et al. 2017; Wan et al. 2021). Similarly, methicillin resistance in *S. aureus* is coded in mobile genetic elements called staphylococcal cassette chromosome *mec* (SCC*mec*), where the methicillin-resistant gene *mecA* is present (Haaber et al. 2017) that codes for the penicillin binding protein 2a. In the current study, as it can be seen that good number of isolates from buck semen were of coagulase-negative origin, and a considerable proportion carried the *mecA* as well as *mecC* gene with some of them detected as phenotypic MRSA (Hanssen and Ericson Sollid 2006). Although its effect on spermatozoa has been well established with spermicidal activity, reduction in sperm quality, but the post-thaw effect of these MRSA especially during artificial insemination on breeding does is unknown. Moreover, the animal handlers who collect the semen, the artificial vagina and other equipments including the vaginal speculum need to be assessed for the presence of MRSA and MRCoNS, including the possibility of horizontal transmission of AMR and other mobile genetic elements from human–animal interaction.

The *mecC* gene is another important homologue of *mecA* gene and found associated with several MRSA lineages including the most common lineage ‘CC130’ (Paterson et al. 2014; Gómez et al. 2021). *mecC* has been often associated with domestic animals (Gómez et al. 2014) including livestock like cattle and lactating cows (Monecke et al. 2007; Schlotter et al. 2014) as well as wild animals (Loncaric et al. 2013), was considered to be zoonotic (Becker et al. 2014). Moreover, the presence of *mecC* positive isolates in the current study especially in the reproductive tract of small ruminants could be treated as a concern that needs immediate control measures. Notwithstanding, it is also important to study the lineages of these current isolates bearing the *mecC* or *mecA* to associate their origin and their transmission dynamics in the future.

Another important resistance is the vancomycin resistance, which was typed using the *vanA* and *vanB* genes. Vancomycin, a glycopeptide antibiotic is the drug of choice for treating patients complicated by MRSA infection, but the last few decades witnessed the development of intermediate (McGuinness et al. 2017) as well as completely resistant strains to vancomycin (Chang et al. 2003; Mukherjee et al. 2021). The current findings were similar to the MRSA genotyping with almost equal number positive for *vanA* and *vanB* in both *S. aureus* and CoNS isolates. The genotyping PCR gives first-hand information regarding the potential AMR characteristics of the isolates in a given entity. In addition, these special genes might express given the apt environment leading to more damage by

horizontal transfer of these genes to highly pathogenic isolates of *Staphylococcus*.

Generally, the prevalence of these bacteria is noteworthy as they may adhere to spermatozoa and decrease sperm motility (Prieto-Martínez et al. 2014). They can also damage acrosomes by producing toxin and cause DNA damage by producing reactive oxygen species (Morrell 2006). Eventually, this could result in decreased fertility rates and culling of breeding animals, and thus causing considerable financial losses to dairy industry (Moustakas et al. 2010). Hence, it is imperative to meticulously preclude the microbial contaminants from entering semen processing and storage facilities and AI operations (Eaglesome and Garcia 1997).

Conclusion

Artificial insemination is a very important tool in assisting reproductive technologies in livestock. The quality of the semen with respect to phenotypic and other functional analyses is well taken care of, while the microbial quality is a very important aspect that needs to be thoroughly addressed. Hence, in this current study, we did the speciation and antimicrobial-resistance (AMR) analysis of staphylococcus isolates from buck semen. The AMR patterns could help decipher important information about the selection of antibiotics in semen dilutor and in turn could help assess the public health risk due to antimicrobial resistant isolates and their influence on personnel assisting the semen collection and processing for AI.

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Declarations

Conflict of interests The authors declare no conflict of interest.

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